Process Characterization Plan

A-Mab Drug Substance — Ultrafiltration / Diafiltration (Step 10) — PCP-010

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AMV, analytical method validation; CQA, critical quality attribute; CPP, critical process parameter; DF, diafiltration; DS, drug substance; DV, diavolume; GPP, general process parameter; HMW, high molecular weight; icIEF, imaged capillary isoelectric focusing; KPP, key process parameter; MSAT, Manufacturing Science and Technology; NOR, normal operating range; PAR, proven acceptable range; RPN, risk priority number;

SEC, size exclusion chromatography; SOP, standard operating procedure; TFF, tangential flow filtration; TMP, transmembrane pressure; UF, ultrafiltration; UV, ultraviolet; WC-CPP, well controlled critical process parameter.

1 Purpose and scope

Ultrafiltration and diafiltration is the last operation of the A-Mab drug substance process. It concentrates the filtrate of the small virus retentive filtration step, exchanges it into the formulation buffer and delivers the drug substance at its target concentration of 75 g/L. This plan defines the characterization study for that operation. It states which parameters will be studied, over which ranges, on which scale-down system, and how the outcome will be judged. It is written before the study is executed and it reports no results.

The study covers all 3 operating parameters of the step, each assessed one at a time over the range given in Table 6.1. Membrane selection, formulation buffer composition and the drug product operations that follow the step are outside the scope of this plan and of the drug substance characterization package. No quality attribute in the drug substance register is assigned to this step, and no viral clearance or impurity clearance is claimed for it. The step is characterized because it sets the concentration and the buffer of the drug substance, and because its operating ranges have to be justified before the process is transferred to the receiving site, Novacyte Biologics — Grafton, WI (Commercial DS).

The plan is written under the lifecycle approach to process validation (U.S. Food and Drug Administration 2011) and the enhanced development approach of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023b, 2012). Its content belongs to Stage 1, process design. The A-Mab case study treats this operation as part of the formulation and reports it with the drug product (CMC Biotech Working Group 2009), and the position taken here is the same in substance. The step is included in the drug substance package for completeness of the transfer, not because a drug substance attribute depends on it.

2 Objectives

The study has five objectives.

1. Confirm that the size variant and charge variant content of the pool is unchanged across the step, over the ranges studied and within the precision of the methods used.
2. Establish the range of each parameter over which product quality remains acceptable and the operation performs as intended, with the other parameters at their set-points.
3. Confirm that the step delivers drug substance at the target concentration, and that the number of diavolumes applied in each run is verified against the mass of permeate collected.
4. Classify each parameter under SOP-4001, with the evidence on which the class rests.
5. Provide the operating ranges, in-process controls and monitoring that the control strategy requires, and the process understanding the receiving site needs to run the step.

3 Related documents

This plan sits under the transfer plan, the master plan and the risk assessment, and it produces one report. Those documents are listed in Table 3.1.

Table 3.1: Documents related to this plan.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-010	Process Characterization Report (Ultrafiltration / Diafiltration (Step 10))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The procedures and validated methods that govern the study are listed in Table 3.2. The controlled document numbers in this corpus are placeholders.

Table 3.2: Controlled procedures and validated analytical methods for the step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3019	Protein Concentration by A280 (UV)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit operation description and prior knowledge

Tangential flow filtration is a platform operation at Novacyte Biologics. It has been used for three earlier humanized IgG1 antibodies (X-Mab, Y-Mab and Z-Mab) and throughout A-Mab clinical manufacture, on the same membrane chemistry and the same operating sequence (CMC Biotech Working Group 2009). The study bounds ranges that are already in routine use. It is not expected to discover new behaviour, and it is not designed to. The type of study the step receives follows from the risk assessment and is given in §4.3.

The operation runs in three phases, of which the first concentrates the feed against an ultrafiltration membrane whose nominal cut-off retains the antibody and passes the buffer species (salts, small ions, excipients of the preceding buffer and other low molecular weight solutes). The retentate is then diafiltered at constant volume against the formulation buffer, so that the species present at the start are washed out exponentially with the number of diavolumes applied. A final concentration phase brings the pool to the drug substance target, after which the retentate is recovered from the system by a buffer flush. Platform experience gives a step yield of about 96.0% with a coefficient of variation of 1.5%. The product that is not recovered is held up in the membrane and in the flow path, and the study does not sample that volume, so it makes no statement about the fate of the material left behind. What it does establish is whether the material that is recovered has changed, from the comparison of final pool against feed described in §5.4.

Two mechanisms could change product quality across the step, and both are the reason the study is run. High transmembrane pressure raises the flux, and with it the degree of concentration polarization at the membrane surface, where the protein is exposed to shear and to an interface, while a high final concentration raises the frequency of self association in the bulk. Both act on the size variant distribution, which is why aggregate is the attribute followed most closely here. The charge variant distribution is followed as well, because the pool is held at its highest concentration during the last phase of the operation and deamidation depends on time.

4.2 Quality attributes in scope

No drug substance quality attribute is formed or cleared at this step. Three attributes are nevertheless in scope, because the pool carries them into the drug substance and the operation could in principle change two of them. They are given in Table 4.1 with their acceptance criteria, their criticality and the Tool #1 score (impact × uncertainty) from which the criticality level follows.

Table 4.1: Drug substance quality attributes carried into the step.

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4
Residual DNA	process impurity	0–0.001 ng/dose	VL	6

Aggregate is the attribute at risk. It is of high criticality, its acceptance criterion is an upper limit, and both of the mechanisms described in §4.1 act on it. Acidic charge variants are of very low criticality and have a wide acceptance range, so they are measured as confirmation and not as a limit the step is expected to approach. Residual DNA is in the table for a different reason. Its acceptance criterion is expressed per dose, and this step brings the drug substance to the concentration on which a dose is based, but the membrane retains residual DNA together with the antibody, so concentrating the pool changes neither quantity relative to the other. The amount per dose is set by the clearance achieved upstream (PCR-005, PCR-007 and PCR-008).

Every attribute in the register is set at an earlier step: Production Bioreactor, Protein A Chromatography, Low-pH Viral Inactivation and Anion Exchange Chromatography. Each is characterized in the plan and report pair for the step that sets it, and the clearance credited to the purification steps is consolidated in PCMR-001. This step is credited with none of that clearance and takes no part in the viral safety argument (International Council for Harmonisation 2023a). What it must show is that it does not undo the work of the steps before it.

4.3 Risk-based prioritization of parameters

RA-001 carried all 3 parameters of this step into the pre-characterization assessment and assigned every one of them to univariate study. The rows are given in Table 4.2.

Table 4.2: RA-001 pre-characterization risk rows for the step.

Parameter	Prospective failure mode	Effect	Sev.	Occ.	Det.	Initial RPN	Assigned study	Priority
Number of diavolumes	Diavolumes < 7	Incomplete buffer exchange (formulation; not part of DS characterization)	4	4	6	96	univariate	Low

Parameter	Prospective failure mode	Effect	Sev.	Occ.	Det.	Initial RPN	Assigned study	Priority
Transmembrane pressure	TMP outside 10-15 psi	Affects flux/processing (formulation)	4	4	6	96	univariate	Low
Final DS concentration	Final concentration outside 65-85 g/L	DS concentration spec (formulation)	4	4	6	96	univariate	Low

All 3 parameters carry the same severity score of 4, which the RA-001 scale calls slight. Their failure modes act on buffer exchange, on flux and on the concentration of the delivered drug substance, and none of those is a quality attribute. Occurrence is scored 4, which the scale calls low, on the ground that the skid controls each of them directly. Detection is scored 6, the score the scale gives to a failure that is seen after the operation and before release, and not during it. The initial risk priority number is 96 for each parameter and each was ranked at low priority.

Two consequences follow for the design of this study. RA-001 assigned no parameter of the step to a designed experiment, and the scores above are the whole basis of that assignment. The ranges must nonetheless be bounded, because the concentration of the drug substance is confirmed by release testing and this step is the last chance to set it. The parameters are studied one at a time, over ranges wider than the ranges in which the step will routinely run.

5 Materials and methods

5.1 Scale-down model and its qualification

The study will be run on a scale-down tangential flow filtration system that reproduces the commercial operation, and no study run may start before that system has been qualified under SOP-1001. The scale-independent parameters are those in Table 6.1, and they are operated at the values proposed for commercial manufacture. The scale-dependent quantities are membrane area, crossflow rate and hold-up volume. These are set so that the small system matches the commercial system on the quantities that govern the operation: membrane chemistry and channel geometry, protein loading per unit membrane area, and crossflow per unit membrane area.

Qualification compares the small system with commercial and pilot data on flux, on process time per diavolume and on the quality of the pool. Step yield is recorded at both scales but is not one of the quantities compared, because hold-up volume makes recovery scale dependent. The acceptance criteria for that comparison are defined in SOP-1001 and are not restated here. The comparison is made at the set-point condition, which is the condition at which the two scales are expected to agree most closely, and it is repeated after any change to the membrane lot or the flow path.

One difference between the scales cannot be removed. The hold-up volume of a small system is a larger fraction of the batch than at commercial scale, so recovery from the final flush is not directly comparable between the two. Step yield at small scale will therefore be reported as an indicator of consistent operation and will not be used to set the commercial yield expectation. This limitation is carried as a study risk in §12. The skid used for the study (EQ-TFF-142) is under calibration and change control, and its calibration status at execution will be recorded in PCR-010.

5.2 Operation

The step is operated under SOP-2013. The feed is the filtrate of the small virus retentive filtration step, which is characterized in PCP-009 and PCR-009. Feed material for the study is drawn from runs of the preceding steps at their set-points, so that the input state of each attribute in Table 4.1 is representative of routine operation.

Each run follows the commercial sequence. The feed is concentrated to the concentration at which diafiltration begins, diafiltered at constant volume against formulation buffer for the number of diavolumes under study, concentrated to 75 g/L unless final concentration is itself the parameter being varied, and then recovered by a buffer flush. Transmembrane pressure is controlled by the skid throughout the concentration and diafiltration phases. The diavolume count is controlled by the recipe and confirmed against the mass of permeate collected.

Materials that are controlled to platform specification are identity checked and released before use, and are not study variables. These are the membrane cassettes, the formulation buffer and the flush buffer. A single membrane lot will be used for the study wherever the schedule allows, so that a lot change cannot be confounded with a parameter effect.

5.3 Analytical methods

The study is supported by 3 validated methods, each listed with its validation reference in Table 3.2. Aggregate is measured by size exclusion chromatography (AMV-3011) and reported as the percentage of high molecular weight species. Charge variants are measured by imaged capillary isoelectric focusing (AMV-3013) and reported as the acidic variant percentage. Protein concentration is measured by ultraviolet absorbance (AMV-3019), which is also the method that decides whether a run met the concentration target.

Table 5.1: Validated performance of the concentration method.

Method	Title	Intermediate precision	LOQ	Accuracy (%)	Variance share
AMV-3019	Protein Concentration by A280 (UV)	1.2 %RSD	0.5 g/L	99.6	0.22

Table 5.1 gives the validated performance of the concentration method. Its intermediate precision is 1.2 %RSD and its limit of quantitation is 0.5 g/L, so it resolves differences that are far smaller than the width of the range studied for that parameter. The validated performance of the size variant and charge variant methods is recorded in the validation reports for AMV-3011 and AMV-3013, and is not reproduced here. The precision of those two methods matters to this study, because it sets the smallest change in an attribute that can be attributed to a parameter rather than to the assay (§5.5).

5.4 Sampling plan

Each run is sampled at four points: the feed, the retentate at the end of diafiltration, the final pool after the last concentration phase, and the permeate. The feed sample establishes the input state of every attribute in Table 4.1 for that run, and the comparison that answers the first objective is the final pool against the feed of the same run. The retentate sample records the protein concentration at which diafiltration ended. The permeate is assayed for protein by ultraviolet absorbance to confirm that the membrane retained the antibody.

Aggregate and charge variants are measured on the feed and on the final pool. Protein concentration is measured on all four samples. Retains of every final pool are held under the conditions defined in SOP-2013, so that an unexpected result can be re-tested on the original material instead of on a repeat run.

5.5 Statistical methods

The planned analysis is descriptive. Each condition is compared with the set-point runs executed on the same feed, and each result is compared with the acceptance criteria in §7. A difference between a condition and the set-point runs is reported as real only when it exceeds the intermediate precision of the method that measured it and the scatter of the set-point runs themselves. Each edge condition is executed once (§6.2), which is too few for a significance test, and none is planned. The number of runs behind every comparison the report makes is stated with the comparison.

No designed experiment is run at this step and no model is fitted to its data. At the steps that do carry a designed experiment, the screening design identifies which parameters matter and the response surface model is the predictive model on which the design space rests. Neither exists here. The outcome of this study is a set of univariate acceptable ranges and a classification for each parameter. It is not a multivariate design space, and this plan claims nothing more for it.

Commercial-scale capability is not estimated from this step. Capability for the drug substance attributes is estimated by Monte-Carlo simulation of 2,000 batches from the qualified models of the steps that set those attributes, and the result is consolidated in PCMR-001. This step contributes the concentration at which the drug substance is expressed, and nothing else, to that calculation.

6 Study design

6.1 Factors, ranges and study type

The design is univariate. Each of the step's 3 parameters is studied across a range wider than the range in which the step will routinely operate, with the other parameters at their set-points. Table 6.1 gives the set-point, the range studied, the normal operating range and the study type for each.

Table 6.1: Parameters, ranges and study type for the step.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Number of diavolumes	DV	8	7–10	7–9	univariate
Transmembrane pressure	psi	12	10–15	10–14	univariate
Final DS concentration	g/L	75	65–85	70–80	univariate

The number of diavolumes is studied over 7–10 DV against a normal operating range of 7–9 DV. The range is extended upward only. Buffer exchange improves as diavolumes are added, so the risk lies at the low edge, and the upper edge is studied to show that the additional process time (the pool spends longer in the system, at its highest concentration) does not change the product. The low edge of the range studied is the low edge of the normal operating range, so this study does not bound an excursion below it. Applying fewer diavolumes than that is prevented by the recipe and detected by the permeate mass check described in §5.2. Completeness of the exchange itself belongs to the formulation and is not characterized here (Table 4.2), and both bounds will be stated again in PCR-010.

Transmembrane pressure is studied over 10–15 psi against a normal operating range of 10–14 psi. The range is extended above the normal operating range because the risk lies there. Shear and concentration polarization both rise with pressure. Below the normal range the consequence is a longer process time and not a change in product quality, which is why the low edge of the studied range and of the normal operating range are the same.

Final drug substance concentration is studied over 65–85 g/L against a normal operating range of 70–80 g/L and a target of 75 g/L. This range is extended in both directions, because both edges carry a consequence. The low edge delivers drug substance below the concentration the process is required to reach, and the high edge is where self association is most likely. The range studied extends beyond the normal operating range on both sides of the target, which follows the practice applied to the characterized steps of this process (PCMP-001) and the enhanced development approach of the case study (CMC Biotech Working Group 2009).

6.2 Univariate assessment

Each parameter is taken to the low and high levels of its characterization range with the other parameters at their set-points, and the set-point condition itself is executed once as the reference for each parameter. The study is 9 runs in total: 6 runs at an edge and 3 at the set-point, with the runs for a given parameter executed on the same feed. No edge condition is replicated, so the scatter of the set-point runs is the only estimate of run-to-run variability the study produces, and §5.5 states how that scatter is used. The schedule is given in Appendix A. This design was chosen over a factorial for the reason given in §4.3, and the choice is a decision about scope and not about effort. A factorial would have to be justified against the risk RA-001 recorded for the step, and that assessment assigned every parameter to univariate study instead.

The assessment is conditional. It establishes how the step behaves when one parameter is at an edge and the others are at their set-points, and it does not establish how the step behaves at a corner where two parameters are at their edges together. That limitation is accepted here, and it is the reason the control strategy will propose the normal operating ranges as the routine limits for the step, rather than the wider ranges studied.

7 Acceptance and decision criteria

Four criteria decide the outcome of the study, and all four are fixed before any data are seen.

The first is the scale-down model. The model is qualified when the comparison required by SOP-1001 is complete and meets the acceptance criteria stated there. Data generated on an unqualified system are not reportable, and the runs affected will be repeated.

The second is product quality. At every condition, aggregate and acidic charge variants in the final pool must be within the drug substance acceptance criteria in Table 4.1. In addition, the change across the step is assessed: a difference between the final pool and its feed that exceeds the intermediate precision of the method is reported as a real change, investigated, and carried into the classification of the parameter that produced it.

The third is process performance, and it is stated separately for the two kinds of run. A run at the set-point condition must deliver a final pool within the normal operating range of the final concentration parameter (70–80 g/L). Its step yield must agree with the yield of the other set-point runs of the study, within the run-to-run scatter recorded when the system was qualified. That comparison stays inside the small scale, for the reason given in §5.1. A run at a deliberately varied condition must complete, and the condition intended for it must be achieved and verified: the mass of permeate collected for the number of diavolumes, the skid record for transmembrane pressure, and the concentration assay for the final concentration. Such a run is judged on product quality and on execution, and not on whether it reached the target concentration.

The fourth is the decision rule for a parameter range. The acceptable range of a parameter is the widest contiguous span of the levels studied at which the product quality criterion is met and the run was executed at its intended condition, with the other parameters at their set-points. The process performance criterion enters through the set-point runs, which fix the reference the levels are compared against. Where every level studied meets those criteria, the whole characterization range is reported as acceptable, and the normal operating range sits inside it with margin on both sides.

Classification follows from the same evidence, under SOP-4001. A parameter with a demonstrated effect on a drug substance quality attribute is classified as a critical process parameter (CPP), or as a well controlled critical process parameter (WC-CPP) where the control capability makes an excursion unlikely. A parameter with no demonstrated effect on a quality attribute, which must nonetheless be held for the process to perform consistently, is classified as a key process parameter (KPP). A parameter with neither property is a general process parameter (GPP). The class is an outcome of the study and is not pre-judged in this plan.

A result that fails one of these criteria is not a failure of the plan. It is recorded, investigated under the site quality system, and reported in PCR-010 together with its effect on the ranges proposed for the control strategy.

8 Proven acceptable ranges (planned analysis)

The report will state a proven acceptable range (PAR) for each parameter, against the two attributes measured across the step, which are aggregate and acidic charge variants. Residual DNA appears in Table 4.1 because the step fixes the concentration on which a dose is based, and it is not assayed in this study, so no range is stated against it (§4.2). The acceptance basis is the drug substance specification, which is the set of criteria in Table 4.1; those are the criteria the drug substance is released against. No viral clearance attribute is in scope at this step, so no step-level clearance contribution has to be back-calculated from a cumulative requirement, as it is for the steps that carry a clearance claim.

The analysis has one form here and not two. Each parameter is evaluated at the levels at which it was actually run, with the other parameters at their set-points, and the PAR is the span of those levels over which the monitored attributes remain within their acceptance criteria. The second analysis used at the designed steps has no counterpart at this one. There the remaining parameters are varied within their normal operating ranges by Monte-Carlo simulation of the fitted response surface model, and here there is no fitted model to propagate. The consequence is stated plainly in the report: every range it gives is conditional on the other parameters sitting at their set-points.

The ranges will be reported as a table of the levels run against the attributes measured, with the acceptance limit for each attribute stated alongside, so that a reader can see the margin between the measured value and its limit at every level executed. The shaded acceptable-region plots used in the reports for the characterized steps are not available here, for the same reason: they are drawn from a fitted model. Two bounds therefore apply to every PAR this study can produce. It is bounded by the levels actually executed, and it is bounded by the assumption that the other parameters are at their set-points. Neither this plan nor its report will describe the result as a design space.

9 Data management and integrity

Study data are recorded contemporaneously in the electronic batch record for the scale-down run and in the validated laboratory systems that hold the analytical results. Records follow the ALCOA+ principles: attributable, legible, contemporaneous, original, accurate, complete, consistent, enduring and available. Instrument output is retained in its original form with audit trails enabled.

Every data set used in PCR-010 is subject to second-person review before it is used. The review confirms that the run was executed at the intended condition, that the sample identity is traceable, and that the analytical result was accepted by the method's own system suitability criteria. Any departure from this plan is documented under the site quality system at the time it happens and reported in PCR-010.

10 Roles and responsibilities

The functions responsible for the study and its deliverables are given in Table 10.1. The approval record at the front of this plan names the functions that approve it.

Table 10.1: Roles and responsibilities for the study.

Function	Responsibility
Process Development / MSAT	Owns the plan, qualifies the scale-down system, executes the runs and authors PCR-010
Analytical Development	Performs the validated methods, releases the analytical data and owns method performance
Manufacturing Science	Confirms that the study conditions represent commercial operation and receives the ranges
Statistics	Reviews the analysis and the comparisons made against the reference condition
Quality Assurance	Approves the plan and the report, and owns the handling of any departure from either

11 Deliverables and schedule

The deliverable of this plan is one report, PCR-010, which contains the executed conditions, the analytical results, the acceptable range of each parameter, the classification of each parameter and any departures from this plan. PCR-010 rolls up into PCMR-001.

The work runs in four phases, in order: qualification of the scale-down system under SOP-1001; execution of the runs listed in Appendix A; analysis and classification under SOP-4001; and issue of PCR-010 for approval. No calendar dates are given here. The study is scheduled within the campaign covered by PCMP-001, and the only sequence constraint that belongs to this step is that its feed comes from the preceding step (§5.2).

12 Risks and assumptions

Three risks could affect the conclusions of this study, and each has a defined response.

The scale-down system may not represent the commercial system in recovery. Hold-up volume is proportionally larger at small scale, and it acts directly on the step yield. The response is in §5.1: yield at small scale is reported as an indicator of consistent operation and is not used to set the commercial expectation.

The feed may not be representative. If the material available for the study differs from routine post-filtration pool in its aggregate content, the comparison of pool against feed within a run remains valid, but the absolute levels reported are not those of a routine batch. The response is to draw the feed from set-point runs of the preceding steps (§5.2) and to report the measured feed state for every run.

Assay variability may mask a small change. The intermediate precision of a method sets the smallest change that can be attributed to a parameter, and a change smaller than that is reported as lying within assay variability and not as an effect (§5.5). This risk is greatest for aggregate, which is the attribute the study is most concerned with and which is present at low levels.

The plan makes three assumptions. Platform behaviour transfers from the earlier antibodies (X-Mab, Y-Mab and Z-Mab) to A-Mab, because the membrane chemistry, the operating sequence and the molecule class are the same. The parameter list in Table 6.1 is complete for this step, which is the position RA-001 reached. The attributes carried into the step are within their acceptance criteria when it begins, which is established by the plans and reports for the steps before it and not by this study.

13 Approvals

This plan takes effect when the functions listed in the approval record at the front of the document have approved it. Approval signifies agreement with the study design in §6, the acceptance and decision criteria in §7 and the planned analysis in §8. A change to any of these after approval requires a revision of this plan, or a documented departure reported in PCR-010.

14 Appendix A — Planned univariate evaluation schedule

The schedule below gives the levels at which each parameter will be run. The other parameters are held at their set-points for every level of the parameter under study, and the reference column is the condition against which the levels of that parameter are compared. That reference condition is the same for every row, and it is executed once for each of them, so those runs are replicates of a single condition and the resulting run count is given in §6.2. The table contains no results.

Table 14.1: Planned evaluation levels, with the reference condition and the normal operating range.

Parameter	Unit	Low level	Reference (set-point)	High level	NOR
Number of diavolumes	DV	7	8	10	7–9
Transmembrane pressure	psi	10	12	15	10–14
Final DS concentration	g/L	65	75	85	70–80

References

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